INCH-POUND

MIL-L-82788(OS)
5 February 1990
SUPERSEDING
WS 20638A
15 July 1986
(See 6.6)

MILITARY SPECIFICATION

LINER, FORMULATION OF (UF-2155)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines one type of rocket motor liner, referred to herein as the liner (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-T-82796	Trimesoyl 1-(-2 Ethyl) Aziridine
MIL-P-82809	Polymer, HTPB, with Antioxidant

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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MIL-C-82813	Curing Agent, Isophorone Diisocyanate
MIL-I-82819	Insulation, Thermal, Polyisoprene, Silica and Asbestos Filled
MIL-C-82829	Carbon Black
MIL-I-82838	Insulation, Thermal - Ethylene Propylene Diene Monomer (EPDM), Cloroprene Rubber, Silica Filled
MIL-A-82843	Adhesive, Elevated Temperature Curing
MIL-S-82858	Silicon Dioxide, Microfine
MIL-P-82859	Propellant, Sustain Grain, (TP-H1205C) (U)
MIL-P-82860	Propellant, Boost Grain (TP-H1206C) (U)

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, Attn: Code 1053, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

ASTM D 846 Ten-Degree Xylene

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 **Manufacture.**

3.2.1 **Materials.** Materials and composition of the liner shall be as specified in tables I and II.

TABLE I. Materials and composition.

Material	Specification	Formulation by Weight	Parts by Weight
Polymer, HTPB, with Anti-oxidant	MIL-P-82809	62.71 1/	
Curing Agent, Isophorone Diisocyanate	MIL-C-82813		
Curing Agent, Trimesoyl, 1-(2-ethyl) Aziridine	MIL-C-82796	3.39	100
Carbon Black	MIL-C-82829	33.90	
Silicon Dioxide, Microfine	MIL-S-82858	as required 2/	0.1 to 6.0

1/ NCO/OH equivalents ratio (see 6.4) can vary from 1.1:1 to 1.6:1. The nominal NCO equivalent value of 0.895 equivalent/100 g (IPDI) and the nominal OH equivalent value of 0.075 equivalent/100 g (liquid polymer) shall be used for the calculation.

2/ Variation of silicon dioxide content of 0.1 to 6.0 parts, based upon 100 parts of other ingredients, allowed for control of rheology for the cure coat liner. The silicon dioxide content for the wet coat liner shall be 0.1 part.

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TABLE II. Raw material weighing data.

Raw material net weights	Maximum deviation allowable
<1 pound (lb) 1/	0.1 g
>1 - 25 lbs	0.01 lb
>25 - 175 lbs	0.1 lb
>175 - 500 lbs	0.25 lb
>500 lbs	0.5 lb

1/ If less than 1 lb weigh in grams (g).

3.2.2 **Weighing.** The weighing of raw materials shall be in accordance with table II. All scales and balances used in weighing of materials shall be calibrated.

3.3 **Liner standardization and verification.** Before any new lot of liner raw material is introduced into liner manufacture, the liner shall be standardized. The standardized liner shall then be subjected to verification tests. The standardization test shall consist of the selection of the NCO/OH ratio in small mixes (400 grams minimum). Verification of the standardized liner shall be accomplished with a minimum of one full scale cure coat mix. Evaluation of the standardization and verification mixes shall be based upon the following characteristics.

3.3.1 **Liner viscosity.** The viscosity of the liner shall be recorded.

3.3.2 **Liner flow characteristics.** The liner shall not flow for the standardization mixes and shall exhibit full coverage for the verification mixes.

3.3.3 **Liner/propellant 90-degree peel strength.** The liner shall exhibit a cured liner to cured propellant 90 degree peel strength exceeding 8 pounds-per linear inch (pli) when prepared and tested as specified in 4.7.4.

3.4 **Shelf life.** The shelf life of the frozen liner shall not exceed 90 days when stored as specified in section 5.

3.5 **Workmanship.** Workmanship shall be such that the liner is uniform in appearance and free from visible contamination.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: room ambient

b. Atmospheric pressure: normal ground

c. Humidity: 90 percent relative, maximum

4.4 First article inspection. First article inspection of the liner shall consist of the examination and tests specified in table III and the quality conformance inspection (see 4.5). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.5. Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests specified in table IV.

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TABLE III. First article inspection.

Examination and Tests	Requirement	Test methods
Visual examination	3.4	4.7.1
Viscosity	3.2.1	4.7.2
Liner flow characteristics	3.2.2	4.7.3
Peel strength	3.2.3	4.7.4

TABLE IV. Quality conformance inspection.

Examination and Tests	Requirement	Test methods
Viscosity	3.3.1	4.7.2
Verifications	3.2, 3.3, 3.4 & 3.5	4.7.5

4.5.1 **Sampling.** A sample of sufficient size to provide all required test specimens shall be taken from each mix of liner.

4.5.2 **Classification of characteristics.** The characteristics verified by the tests and examination herein are classified as Major or Minor in accordance with WR-43. Tests and examinations that verify major characteristics are identified with the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. When the classification does not apply to the complete paragraph, the portion to which the classification is applicable is enclosed in brackets. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 **Standardization test batch preparation.** Standardization test batches shall be prepared using materials and composition as specified in table I using a minimum of three NCO/OH ratios. The nominal microfine silicon dioxide level for the cure coat shall be 3.5 parts. The silicon dioxide level for the wet coat liner shall be 0.1 part.

4.7 **Methods of inspection. The following inspection methods shall be** used. Unless otherwise specified in the test or procedure description, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

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4.7.1 **Visual examination.** When compounded into liner as specified in 3.1, the liner material shall be visually examined for conformance to 3.4.

4.7.2 **Viscosity test.** Each batch mixed shall be tested for viscosity, within 5 minutes after the end of mix, using the HBT Brookfield viscometer with a TA spindle at 5 revolutions per minute (rpm).

4.7.3 **Liner flow characteristics.** Standardization mixes of cure coat liner shall have liner applied to steel plates to a thickness range of 0.02 inch to 0.060 inch. When subjected to 150 ± 5 degrees Fahrenheit (°F) for a minimum of 16 hours, the flow plates representing the design thickness shall conform to 3.3.2. Verification cure coat liner mix, when applied into a case simulator of the approximate diameter of the full scale case, shall conform to 3.3.2.

4.7.4 **Liner/propellant 90 degree peel strength (M101).** The peel test specimens shall be prepared and tested using an Instron tensile tester as follows:

a. From vulcanized sheet stock 0.1 ± 0.02 inches thick of EPDM conforming to MIL-T-82838 and polyisoprene insulation conforming to MIL-I-82819, prepare surfaces by abrading and degreasing. The polyisoprene insulation shall be grit blasted with zirconium silicate and the EPDM shall be abraded with 100 to 150 grit abrasive paper. Both insulations shall be degreased with methyl ethyl ketone (MEK) conforming to ASTM D 740.

b. The polyisoprene insulation shall be coated with adhesive conforming to MIL-A-82843 by spraying 0.25 ± 0.06 gram per square inch of a 50/50 by volume mixture of the MIL-A-82843 adhesive and xylene conforming to ASTM D 846.

c. Dry both the adhesive coated insulation and the EPDM insulation for 60 ± 12 hours at 170 ± 10°F.

d. Cut a minimum of two panels of polyisoprene insulation and a minimum of one panel of EPDM insulation to an approximate size of 4 5/8 x 6 5/8 inches.

e. Apply with template, 11.5 ± 1.2 g of liner to each panel and cure for 32 ± 4 hours at 150 ± 5°F (this liner is the nominal 3.5 part silicon dioxide formulation).

f. Brush apply 2.9 ± 0.2 g of liner to each panel (this liner is the 0.1 part silicon dioxide formulation).

NOTE: The insulation shall be stored and sealed in polyethylene with desiccant from step c through step f if any stop in the process is delayed by more than 4 hours.

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g. Install one polyisoprene panel and one EPDM panel in one 1/2-gallon carton, preheat for 20 to 30 minutes at 135 ± 5°F and cast with propellant conforming to MIL-P-82859 at a casting pressure of 0.2 to 0.5 inch mercury (Hg) absolute pressure. Casting time shall be 3 to 5 minutes.

h. Install one polyisoprene panel in a 1/2-gallon carton, preheat for 20 to 30 minutes at 135 ± 5°F and cast with propellant conforming to MIL-P-82860 at a casting pressure of 0.2 to 0.5 inch Hg absolute pressure. Casting time shall be 3 to 5 minutes.

i. Propellant cure parameters shall be within 4 percent of the nominals established during standardization testing of the corresponding lot of the following propellant:

MIL-P-82859 Propellant, Sustain Grain

MIL-P-82860 Propellant, Booster Grain

j. Prepare peel specimens by cutting 1 inch wide by 6 1/2 inches long by 1 inch deep composite samples from each panel. Bond a strip of wood 1 1/4 x 7 5/8 inches to propellant surface with a suitable adhesive.

k. After adhesive cure, separate liner from propellant for 1 ± 0.2 inch, starting the separation at the end of the sample that was in the bottom of the carton.

l. Condition sample for a minimum of one half hour at 75 ± 5°F. Place sample in test trolley, attach insulation tab end in gripping jaws of Instron tester and test at an angle 90 ± 5 degrees at a crosshead rate of 5.0 inches/minute with the chart speed set at 5 inches/minute.

m. Determine the peel strength by dividing the average force required to maintain a peeling action by the width of the peel sample. The initial one-half inch and the last one inch of crosshead travel shall not be used in the determination.

n. The peel strength for acceptance shall be determined by taking the average of the values as determined in 4.7.4 m of at least three 90-degree peel specimens.

Alternate Method:

a. Prepare 0 ± 0.20 thick by 6 inch X 6 inch vulcanized sheet stock. Two sheets EPDM, four sheets polyisoprene per cure coat liner mix.

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b. Half of one side of the insulation sheet will be abraded with 100- to 150-grit sandpaper. The sheets shall then be cleaned with MEK. (see 4.7.4a).

c. The polyisoprene shall be brush coated on one side with adhesive conforming to MIL-A-82843.

d. Cure adhesive-coated polyisoprene and EPDM parts at ± 10°F for 60 hours ± 12 hours.

e. Apply cure coat liner to the abraded and or adhesive-coated halves of rubber plaques. Coverage is to be 6.5 ± 1 g per plaque.

f. Cure lined plaques for 32 ± 4 hours at 150°F ± 10°F.

h. Plaques are labeled and spit, and two peel boats per plaque are prepared.

i. 4 polyisoprene peel boats, and 4 EPDM peel boats per cure coat liner mix are to be cast with sustainer propellant. Four polyisoprene peel boats per cure coat liner mix are to be cast with booster propellant.

NOTE: Peel boats have a 2" X 2" contact patch between liner and propellant.

j. Cure cast peel boats with established propellant cure parameters.

k. Pull conditioned peel board (70 ± 5° for 1/2 hour minimum) on Instron 1122.

l. Cross head speed - 1.0 ipm
Head setting - 20 psi
Chart speed - 1.0 ipm

m. Peel strength acceptance shall be based on the average of 3 to 90 peels. Peel values must be greater than 8 pli.

4.7.5 **Verifications.** The following verification shall be required:

a. (M102) Examination of the batch mix data to verify that the lots of raw materials used in the batch have met the requirements of 3.2, 3.3 and 3.4.

b. (M103) Examination of the weight records to verify that the actual weights of the raw materials used in composition of the liner are within the limits specified in table I and table II.

c. Examination of the storage records to verify conformance to 3.4.

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4.8 Inspection of packaging. The liner containers and container markings shall be visually examined for compliance with section 5.

5. PACKAGING

5.1 Preservation. The liner material shall be packaged in containers and sealed with lids that will prevent the circulation of atmospheric air within the containers. Liner material that is to be stored frozen shall be packaged in tin-coated, sanitary (food pack) quality steel containers with no resinous internal coating.

5.2 Marking. Containers shall be marked to show the liner material manufacturing date (end of mix), mix number, and shelf life expiration date.

5.3 Handling and storage. Liner material that is to be stored frozen shall be placed in an environment of zero ± 10°F within 6 hours after end of mix and stored at that temperature. Frozen liner material shall be thawed before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The liner covered by this specification is intended for use in the Standard Missile Mk 104 Dual Thrust Rocket Motor chambers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article inspection is required (see 3.1).

d. Inspection conditions, if other than as specified.

e. Production lot size, if other than as specified.

f. Lot numbers, if not as specified.

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.4. The contracting officer

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should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 **NCO/OH equivalents ratio.**

$$\text{NCO/OH} = \frac{(\text{percent IPDI})\ (0.895)}{(\text{percent liquid polymer})\ (0.075)}$$

6.5 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.6 **Supersedure information.** MIL-L-82788 incorporates the following engineering change proposals (ECPs), notice of revision (NOR) and specification change notice (SCN):

ECP		NOR		SCN	
MTA023	11/24/86			SCN 3	11/24/86
ARC002	9/29/88	ARC002.2	9/29/88		

Preparing Activity:
NAVY-OS
(Project 1340-N795)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-L-82788	2. DOCUMENT DATE (YYMMDD) 5 February 1990

3. DOCUMENT TITLE
LINER, FORMULATION OF (UF-2155)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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